

APARAJITA KASHYAP

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EDUCATION

Columbia University Department of Biomedical Informatics	New York, NY
Doctor of Philosophy (PhD)	September 2022-Present
MA, Biomedical Informatics	January 2024
Johns Hopkins University	Baltimore, MD
Bachelors of Arts (BA) in Biophysics	May 2022
<i>Minors:</i> Applied Mathematics and Statistics, Space Science and Technology	

RESEARCH EXPERIENCE

Columbia University	September 2022-Present
Graduate Researcher, Department of Biomedical Informatics	
Advisors: Shalmali Joshi, PhD and Noémie Elhadad, PhD	
Johns Hopkins University	November 2019-August 2022
Undergraduate Researcher, Department of Physics and Astronomy & Department of Biophysics	
Advisor: Brian Camley, PhD	
<i>Departmental Honors Thesis:</i> Optimization of Concentration Sensing in a Dynamic Environment	
Memorial Sloan Kettering Cancer Center	June 2021-August 2021
Computational Biology Summer Intern	
Advisor: Christina Leslie, PhD	
Johns Hopkins University Applied Physics Laboratory	April 2020-June 2021
CIRCUIT Intern	
Supervisor: Hannah P. Cowley, William Gray-Roncal, PhD	
Johns Hopkins University School of Medicine	February 2019-May 2020
Undergraduate Researcher, Department of Otolaryngology	
Advisor: Mark Shelhamer, PhD	

PREPRINTS AND PUBLICATIONS

1. **Kashyap A**, Matijevic S, Elhadad N, Kushner SA, Joshi S. A pipeline for enabling path-specific causal fairness in observational health data. arXiv [preprint]. 2026 January. DOI: 10.48550/arXiv.2601.09841
2. **Kashyap A***, Allsman CJ*, Campbell EA, Desai PM, Volpe SG, Massey BP, Bright TJ, Bakken S, Bear Don't Walk IV OJ, Pichon A. Contextualizing key principles to promote a justice-oriented informatics research agenda: proceedings and reflections from an AMIA workshop. JAMIA 2025. (*In Press*)
3. Alsentzer E, Charpignon ML, Chen B, ... **Kashyap A**... Jeong H, Thakoor K. Reflections from Research Roundtables at the Conference on Health, Inference, and

Learning (CHIL) 2025. arXiv [preprint]. 2025 October 17. DOI: 10.48550/arXiv.2510.15217

4. Pang C, Jeanselme V, Choi YS, Jiang Xs, Jing Z, **Kashyap A**, Kobayashi Y, Li Y, Pollet F, Natarajan K, Joshi S. FoMoH: A clinically meaningful foundation model evaluation for structured electronic health records. arXiv [preprint]. 2025 May 23. DOI: 10.48550/arXiv.2505.16941.
5. **Kashyap A**, Aziz M, Sun TY, Lipsky-Gorman S, Opoku-Anane J, Elhadad N. Investigating Racial Disparities in Drug Prescriptions for Patients with Endometriosis. npj women's health 2025. DOI: 10.1038/s44294-025-00053-3
6. Reyes Nieva H, **Kashyap A**, Voss EA, Ostropolets A, Anand A, Ketenci M, DeFalco FJ, Natarajan K, Choi YS, Li Y, Allen MN, Guang SA, Elhadad N. The Impact of Evolving Endometriosis Guidelines on Diagnosis and Observational Health Studies. medRxiv [preprint]. 2024 Dec 16. DOI: 10.1101/2024.12.13.24319010.
7. **Kashyap A**, Wang W, Camley BA. Tradeoffs in concentration sensing in dynamic environments. Biophysical Journal 2024. DOI: 10.48550/arXiv.2310.00062
8. Cowley HP, Robinette MS, Matelsky JK, Xenos D, **Kashyap A**, Ibrahim NF, Robinson ML, Zeger S, Garibaldi BT, Gray-Roncal W. Using machine learning on clinical data to identify unexpected patterns in groups of covid-19 patients. *Scientific Reports*, 13(1). February 2023. <https://doi.org/10.1038/s41598-022-26294-9>

PRESENTATIONS AND INVITED TALKS

1. **Kashyap A**, Kushner S, Elhadad N, Joshi S. Leveraging data-driven analyses to improve subpopulation robustness of a schizophrenia prediction model. Podium Abstract. AMIA 2025 Annual Symposium. Atlanta, GA; 11/2025
2. **Kashyap A (Speaker)**, Elhadad N, Kushner SA, Joshi S. Leveraging data-driven analyses to characterize subpopulation robustness. *Invited talk*: University of Washington Biomedical Informatics Research Colloquium. Seattle, WA; 10/2025.
3. **Kashyap A (Speaker)**, Elhadad N, Kushner SA, Joshi S. Machine learning predicts diagnostic transition to schizophrenia using administrative claims data. *Invited talk*: SNF Center for Precision Psychiatry and Mental Health. New York, NY; 03/2025.
4. Nong P, El-Azab S, **Kashyap A (Speaker)**, Newman-Griffis D, Bear Don't Walk IV OJ. Making justice and equity tangible in informatics. Panel Speaker. AMIA 2024 Annual Symposium. San Francisco, CA; 11/2024
5. **Kashyap A (Speaker)**. Machine learning for time series data. *Invited Lecture*: Machine Learning for Healthcare. Columbia University, New York, NY; 4/2024
6. Pichon A, Volpe S, **Kashyap A (Speaker)**, Desai P, Anand T, Campbell E, Schiffer K, Diamond C, Massey B, Allsman C, Newbury A, Richter L, Bright T, Bakken S, Bear Don't Walk IV, OJ. Synthesizing gaps and priorities for a justice informatics research agenda. AMIA 2023 Annual Symposium. New Orleans, LA; 11/2023.
7. **Kashyap A (Speaker)**, Aziz M, Sun TY, Lipsky-Gorman S, Opoku-Anane J, Elhadad N. Investigating racial inequality in drug prescriptions for patients with endometriosis. Oral Presentation. World Congress on Endometriosis. Edinburgh, UK; 05/2023.
8. **Kashyap A (Speaker)**, Camley B. Tradeoffs in concentration sensing in dynamic environments. Oral Presentation. Johns Hopkins University Biophysics Undergraduate Research Festival. Baltimore, MD; 05/2022.

POSTERS

1. **Kashyap A**, Elhadad N, Kushner SA, Joshi S. Evaluating intersectional causal fairness for schizophrenia risk prediction. Conference on Health, Inference, and Learning (CHIL). Berkeley, CA; 06/2025.
2. Reyes Nieva H, **Kashyap A**, Voss EA, Ostroplets A, Anand A, Ketenci M, Guang S, Allen MN, DeFalco FJ, Choi YS, Li Y, Natarajan K, Ryan P, Elhadad N. Expanding Endometriosis Diagnostic Guidelines Captures More Women with Non-Classical Symptoms, Different Care-Seeking Patterns, and Lack of Access to Surgical Care. Endometriosis: A Look at Current Practices and Emerging Science, New York Academy of Sciences. New York, NY; 06/2025.
3. **Kashyap A**, Elhadad N, Kushner SA, Joshi S. Leveraging data-driven analyses to improve subpopulation robustness of a schizophrenia prediction model. Columbia University Data Science Day. New York, NY; 04/2025.
4. Desai PD, Raj A, **Kashyap A**, Albers D, Mamykina L. Something Similar: Exploring the Usefulness of an On-the-Spot Meal Recommendation System for Health Goal Attainment. Extended Abstracts of the 2025 ACM SIGCHI Conference on Human Factors in Computing Systems. Tokyo, Japan; 4/2025
5. Kolo A, Pang C, Choi E, Steinberg E, Jeong H, Gallifant J... **Kashyap A**... King RC. MEDS Decentralized, Extensible Validation (MEDS-DEV) Benchmark: Establishing Reproducibility and Comparability in ML for Health. Demo Track, Machine Learning for Health (ML4H). Vancouver, Canada; 12/2024
6. **Kashyap A**, Elhadad N, Kushner SA, Joshi S. Leveraging machine learning explanations to assess the appropriateness of using demographic information during schizophrenia onset prediction. AMIA 2024 Annual Symposium. San Francisco, CA; 11/2024.
7. Reyes Nieva H, **Kashyap A**, Voss EA, Ostroplets A, Anand A, Ketenci M, DeFalco FJ, Natarajan K, Choi YS, Li Y, Allen MN, Guang S, Elhadad N. The Impact of Evolving Diagnostic Guidelines on Clinical Characterization of Endometriosis. OHDSI 2024 Global Symposium. New Brunswick, NJ; 10/2024.
8. **Kashyap A**, Aziz M, Sun TY, Lipsky-Gorman S, Opoku-Anane J, Elhadad N. Investigating racial inequality in drug prescriptions for patients with endometriosis. National Library of Medicine T15 Training Conference. Stanford University; 06/2023.
9. **Kashyap A**, Sun TY, Elhadad N. Data-driven identification of symptoms of interest in the early detection of endometriosis. World Congress on Endometriosis. Edinburgh, UK; 05/2023.
10. Ashiru O*, **Kashyap A***, Koroma F*, Ibrahim N, Liu E, Robinette M, Cowley HP, Gray-Roncal W. Generalizable precision medicine tools for patient cohort discovery and visualization. Johns Hopkins University DREAMS Symposium. Baltimore, USA; 04/2021
11. Tang S, Chandra V, **Kashyap A**, Kilburn W, Spencer C, Yaovatsakul K, Nguyen J, Roberts D, Shelhamer M. Multivariate analysis of human health and performance in an acute spaceflight simulation. NASA Human Research Program Investigator's Workshop. Galveston, USA; 01/2020
12. Tang S, Tan M, Chandra V, **Kashyap A**, Kilburn W, Roberts D, Shelhamer M. Vestibular and postural assessment device and methods. NASA Human Research Program Investigator's Workshop. Galveston, USA; 01/2020

Aparajita Kashyap

* Denotes equal contribution

TEACHING EXPERIENCE

Teaching Assistant: Acculturation to Programming and Statistics

Fall 2023

Teaching Assistant: Machine Learning in Healthcare

Spring 2024

LEADERSHIP AND SERVICE

Public Interest Technology (Barnard College)

Graduate Assistant, January 2025-Present

Developing extracurricular programming about Public Interest Technology for undergraduate students at Barnard College. Planned hackathon that resulted in two student submissions to the NASA Space Apps Challenge (Oct 2025).

Justice Informatics Group

Member, September 2022-Present

Mission: Identify existing methods and research gaps for justice-oriented research in biomedical informatics

Disability Ethics, Intersectionality

& AI/ML Bias Speaker Series

Series Coordinator, January 2023-May 2024

Organizing speaker series and leading discussion groups relating to disability ethics within informatics and AI/ML

Columbia Precision Medicine & Society

Graduate Fellow, November 2023-May 2024

Organizing and leading workshops relating to precision medicine, ethics, privacy, and fairness